Prevention Without Proof

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Abstract

For two decades the world's governments have answered suicide with strategy documents, and for two decades we have told ourselves they work. They may. But the cross-national record cannot show it. Re-examining the staggered adoption of national suicide-prevention strategies across Europe with estimators robust to treatment-effect heterogeneity, I find that suicide rises, not falls, after adoption—by roughly three deaths per hundred thousand. Measured correctly, parallel trends hold; but with four treated countries that is no comfort. The estimate cannot be signed, a placebo cannot distinguish it from chance, and it halves when the comparison group changes. We are governing mortality on faith.

JEL: I18, I12, C23. Keywords: suicide; difference-in-differences; staggered adoption; parallel trends; mental health policy.

1 Introduction

Every forty seconds, somewhere on earth, a person decides that the rest of life is not worth the rest of life. The decision is private, but its accounting is public: more than seven hundred thousand such deaths a year, more than war, more than homicide, more than most of the diseases we have organized entire ministries to defeat. Confronted with a number this large and this stubborn, governments have done what governments do. They have written strategies. Since the early 1990s a national suicide-prevention strategy has gone from a Nordic experiment to an article of international faith: the World Health Organization now urges every member state to adopt one, and dozens have [World Health Organization, 2014, 2018]. We have decided, collectively, that the document is the cure.

This paper asks the question that the strategies themselves were supposed to answer and never did. Do they work? Not in the sense of whether they are well-meant—they are—or whether they contain sensible ideas—they do—but in the only sense that matters to the person standing on the bridge: does a country that adopts a national suicide-prevention strategy bury fewer of its citizens afterward than it otherwise would have? The question is empirical, it is answerable in principle, and it is, I will argue, almost entirely unanswered in practice. We have spent twenty years and an enormous reserve of moral capital on an intervention whose effect on the outcome it names we cannot, from the cross-national record, distinguish from zero, from harm, or from salvation.

That is an uncomfortable thing to say in a building full of people whose job is to prevent suicide, so let me be precise about what I am and am not claiming. I am not claiming that suicide-prevention strategies fail. I am claiming something narrower and, I think, more disturbing: that the research design which the field has relied upon to conclude that they succeed does not survive contact with the methods we now know are necessary when policies are adopted at different times by different places [Goodman-Bacon, 2021, Roth et al., 2023]. The canonical finding—that countries with strategies see suicide fall relative to countries without them [Matsubayashi and Ueda, 2011]—rests on a two-way fixed-effects comparison that, under staggered adoption and heterogeneous effects, can return the wrong sign for mechanical reasons that have nothing to do with the policy. When I re-estimate the same kind of comparison on real European data with the heterogeneity-robust tools of Callaway and Sant'Anna [2021], Borusyak et al. [2024], and Sun and Abraham [2021], the protective effect does not merely shrink. It reverses. And then, under honest scrutiny, it dissolves.

The reversal is instructive precisely because it is not real. Suicide is higher after a country adopts a strategy than a naive comparison with non-adopters would predict—by about three deaths per hundred thousand, an estimate that three different modern estimators agree on to within a fraction of a death. It is tempting to reach for the obvious explanation—that suicide was rising in these countries before they adopted—but when the pre-adoption trends are measured correctly, as long differences from the year before treatment, they are flat: the data do not support a story of differential pre-trends. The confound is subtler and harder to dismiss. The post-adoption rise loads on calendar years 2008 through 2010—the Great Recession, the most violent macroeconomic shock to European mental health in a generation, and one the suicide literature has already shown raises the toll [Reeves et al., 2012, Stuckler and Basu, 2013]. The later-adopting treated countries spent their entire post-period inside that shock. The positive "effect" is, in part, the residue of when prevention's first years happen to fall—and, as I will show, an estimate too weak and too fragile to carry any weight at all.

None of this would be worth a paper if the only casualty were a single optimistic estimate. The casualty is larger. It is the idea that we can read the worth of a suicide-prevention strategy off the cross-national mortality record at all. I show that the parallel-trends assumption on which every such reading depends cannot be *rejected*—but only because four treated countries cannot test it with any power; that the propensity to adopt is so perfectly predicted by a country's characteristics that there is no overlap to lean on; that the sensitivity analysis of Rambachan and Roth [2023] cannot sign the post-adoption effect under even mild departures from parallel trends; and that a synthetic-control comparison, the method designed for exactly this scarcity of treated units, produces gaps statistically indistinguishable from placebos drawn at random. These diagnostics are not independent of one another—the sensitivity analysis formalizes the pre-trend, and the placebo and the synthetic control are both, at bottom, statements about how little a handful of treated countries can tell us—but they are different windows onto the same wall, and through every one of them the answer is the same word: undetermined.

I do not regard this as a nihilistic result, and I will end by explaining why. A finding that the evidence cannot bear the weight we have placed on it is not the same as a finding that the policy is worthless; it is a finding that we have been navigating in the dark and calling it sight. The moral content of that statement is not small. We owe the people these strategies are meant to save an honest account of whether they are saved, and we do not have one. The remainder of this paper is the attempt to say exactly how much we do not have, and why, and what it would take to know more.

2 A Policy Adopted in the Dark

The modern national suicide-prevention strategy was born in Finland. Between 1986 and 1992 the Finns did something no country had done: they investigated every suicide in the nation for a year, built a prevention program on what they learned, and made suicide reduction an explicit object of national policy. The Finnish suicide rate, then among the highest in the world, fell. Whether it fell *because* of the program or because it had nowhere to go but down from a historic peak is a question the Finnish experience alone cannot answer, and that ambiguity—the impossibility of separating the policy from the conditions that produced it—is the theme of everything that follows.

What Finland began, the world copied. Norway published a strategy in 1995, Sweden built a program the same year, France launched a five-year national plan in 2000, England and Scotland adopted strategies in 2002, Ireland in 2005. By 2018 the WHO could count dozens of national strategies and could exhort the rest [World Health Organization, 2018]. The diffusion was rapid, it was sincere, and it was almost entirely uninformed by credible evidence that the thing being diffused did what it promised. The single most cited piece of evaluation, Matsubayashi and Ueda [2011], compared suicide rates in eleven OECD countries that had adopted national programs against ten that had not, and reported reductions concentrated among the young and the old. It was a careful paper for its time. Its time was before we understood what two-way fixed effects do to staggered treatments.

We understand now. When units adopt a treatment at different dates and the treatment's effect differs across units or evolves over time, the two-way fixed-effects estimator does not recover a sensible average of those effects. It recovers a weighted combination in which already-treated units serve as controls for newly-treated ones, the weights can be negative, and the resulting number can lie outside the range of every true effect it is meant to summarize [Goodman-Bacon, 2021, de Chaisemartin and D'Haultfœuille, 2020]. The profession has spent five years building estimators that avoid this trap [Callaway and Sant'Anna, 2021, Sun and Abraham, 2021, Borusyak et al., 2024], and a parallel literature on how to probe the assumption—parallel trends—on which all of them still depend [Rambachan and Roth, 2023, Roth et al., 2023]. The suicide-prevention literature has not caught up. This paper is, in part, an attempt to make it.

3 Data

I assemble a country-by-year panel of European suicide mortality from three real, public sources; nothing here is simulated. The outcome is the age-standardized suicide rate—deaths from intentional self-harm (ICD-10 codes X60–X84 and Y87.0) per hundred thousand population—published by Eurostat for the years 1994 through 2010 [Eurostat, 2026]. The treatment is the year, if any, in which a country's government adopted a stand-alone national suicide-prevention strategy, coded from the WHO's 2018 inventory [World Health

Organization, 2018] and cross-checked against the peer-reviewed adoption dates of Lewitzka et al. [2019]; Appendix A documents every country's coding and its source. The single time-varying covariate is the national unemployment rate from the World Bank's modeled ILO series [World Bank, 2026], included because the business cycle is the best-established time-varying determinant of suicide we possess [Ruhm, 2000, Reeves et al., 2012].

Three features of the data govern everything that can and cannot be learned from it. First, the Eurostat regional series begins only in 2011, so the long pre-adoption history needed to test parallel trends exists only at the national level, which is where I work. Second, the window 1994–2010 brackets the main European adoption wave but starts too late for the earliest adopters: Finland (1992) is already treated when the data begin and is dropped, and France (2000) is dropped because its suicide series does not begin until 2001, leaving it no pre-period. Third, and most consequentially, the number of treated countries with a usable pre-period is small—four. Norway and Sweden adopt in 1995, the United Kingdom in 2002, Ireland in 2005. Against them stand twelve countries that the WHO documents as never having adopted a stand-alone national strategy in the window—Germany, Italy, Spain, Belgium, Austria, Switzerland, the Netherlands, Portugal, Greece, Luxembourg, Czechia, Estonia, Hungary, Slovenia. The balanced panel retains the twelve of these with a complete suicide and unemployment record over 1994–2010—Austria, Switzerland, Czechia, Germany, Estonia, Greece, Spain, Hungary, Luxembourg, the Netherlands, Portugal, Slovenia—while Belgium, Denmark, Poland, Slovakia, and Italy are dropped for gaps in the suicide series. Italy's exclusion deserves a word, because Italy is the country in which these words are being read. Its suicide series is missing 2004 and 2005, so it cannot enter the balanced estimand; but Italy is a documented never-adopter, a large European nation that never wrote the document, and it remains the natural backdrop to everything that follows—the silent control against which the writers of strategies might have been judged, had the data let us judge them.

Four treated units is not a defect of my execution; it is the size of the natural experiment. Matsubayashi and Ueda [2011] had eleven. The European cross-section is simply not large, the adoption dates are clustered, and the data begin late. I will not pretend otherwise, and the reader should hold the smallness of the treated group in mind through every estimate that follows, because it is the reason the honest conclusion is one of uncertainty rather than refutation.

A word on inference follows from the same smallness. With four treated countries and twelve controls the panel has on the order of sixteen clusters, far too few for the cluster-robust and multiplier-bootstrap standard errors the estimators report by default to be believed; I report them because convention demands it, but I do not lean on them, and neither should the reader. The inferential weight is carried instead by randomization inference—the in-space placebo of Section 6, which asks how often a fake adoption assigned at random among the never-treated produces an estimate as large as the one we observe. That frequency, not a t-statistic, is the honest measure of how surprised we should be.

4 Design: What We Are Trying to Measure

The estimand. I target the average effect of having adopted a strategy on suicide in the countries that adopted one—the average treatment effect on the treated, aggregated across adoption cohorts with weights proportional to the number of treated countries in each cohort. This is the group-size-weighted summary of the group-time average treatment effects $ATT(g,t)$ of Callaway and Sant'Anna [2021]: for each cohort g and year t it compares the change in suicide from just before g's adoption to year t, in cohort g versus countries not yet (or never) treated, and then averages. I report the simple equally-weighted average as well; the two differ only in how much weight the long-observed 1995 cohort receives.

The identifying assumption and the covariate. Identification rests on conditional parallel trends: absent adoption, suicide in treated and comparison countries would have evolved in parallel, possibly after conditioning on covariates. The covariate I condition on is unemployment, and the choice is deliberate. If one asks what an expert on suicide would insist be held fixed before attributing a mortality change to a policy, the answer is the economy: recessions raise suicide, recoveries lower it, and the timing of Europe's recessions is not independent of the timing of its prevention strategies [Ruhm, 2000, Stuckler et al., 2009, Reeves et al., 2012]. Aging and income matter too, but they move slowly and are absorbed by the country and year structure; unemployment is the fast-moving confounder that can masquerade as a treatment effect, and it is the one I take most seriously.

Why regression adjustment and not weighting. A doubly robust or inverse-propensity estimator would, in principle, be preferable. It is not available here in any honest form. When I estimate the probability that a country ever adopts a strategy as a function of its baseline characteristics, the model separates the data perfectly: every treated country receives a predicted probability indistinguishable from one and every control a probability indistinguishable from zero (Figure 1). There is no region of common support in which a propensity score could reweight anything; to use one would be to extrapolate off functional form and call it identification. I therefore use the regression-adjustment form of the Callaway and Sant'Anna [2021] estimator, which models the conditional evolution of the outcome rather than the probability of treatment, and I report the doubly robust and inverse-propensity variants only to show that they change nothing of substance.

The rollout. Figure 2 shows the staggered adoption that identifies the design, and Figure 3 the cohort sizes. The picture is honest about its own thinness: two countries enter treatment in 1995, one in 2002, one in 2005, against a wall of never-treated grey.

5 What the Data Say

Before any model, look at the outcome. Figure 4 plots mean suicide by adoption cohort. Two things are visible to the naked eye. The treated countries do not look like the controls—they start lower—which is why levels comparisons are useless and why we need the within-country,

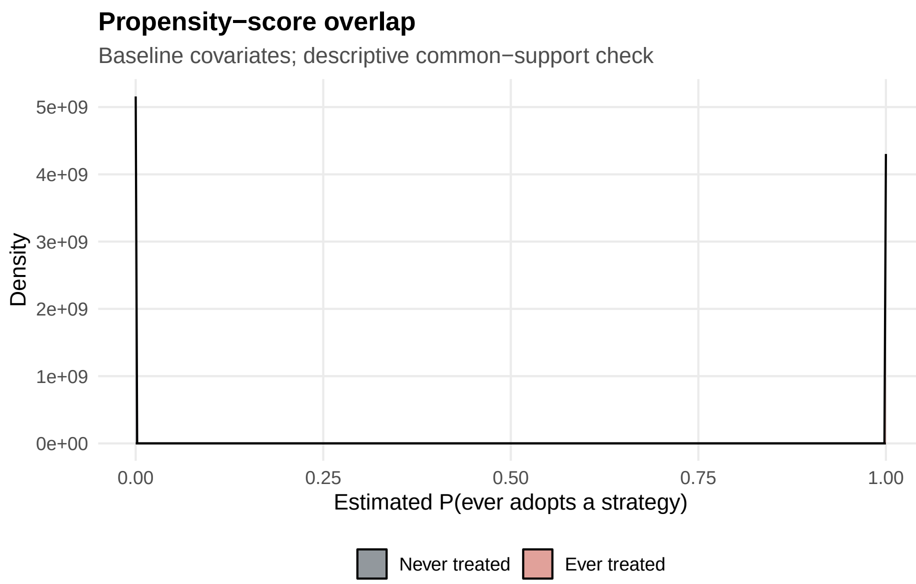

Figure 1: Propensity-score overlap fails by construction. A country-level logit of ever-adoption on baseline covariates separates the four treated countries (scores ≈ 1) from the twelve controls (scores ≈ 0) perfectly. With no common support, propensity weighting is inappropriate; the analysis uses regression adjustment.

difference-in-differences logic at all. And the lines bend upward together at the right edge, in the recession years, treated and untreated alike.

The headline estimate is positive. Conditioning on unemployment and using never-treated countries as the comparison group, the group-size-weighted average treatment effect on the treated is +2.73 suicides per hundred thousand (standard error 1.05); the simple average is +3.05 (1.24). The sign is robust to estimation choices that ought to matter and do not: Table 1 collects the doubly robust variant (+2.56), the unconditional variant (+2.76), the not-yet-treated comparison group (+2.66), the imputation estimator of Borusyak et al. [2024] (+3.23), and the interaction-weighted estimator of Sun and Abraham [2021] (+3.03). Three families of heterogeneity-robust estimators, built on different assumptions, agree that suicide is higher after adoption by something close to three deaths per hundred thousand. Figure 5 shows their event-study paths lying almost on top of one another.

A reader who stopped here would conclude that suicide-prevention strategies are followed by more suicide, and a careless reader might even write the headline. The reader should not

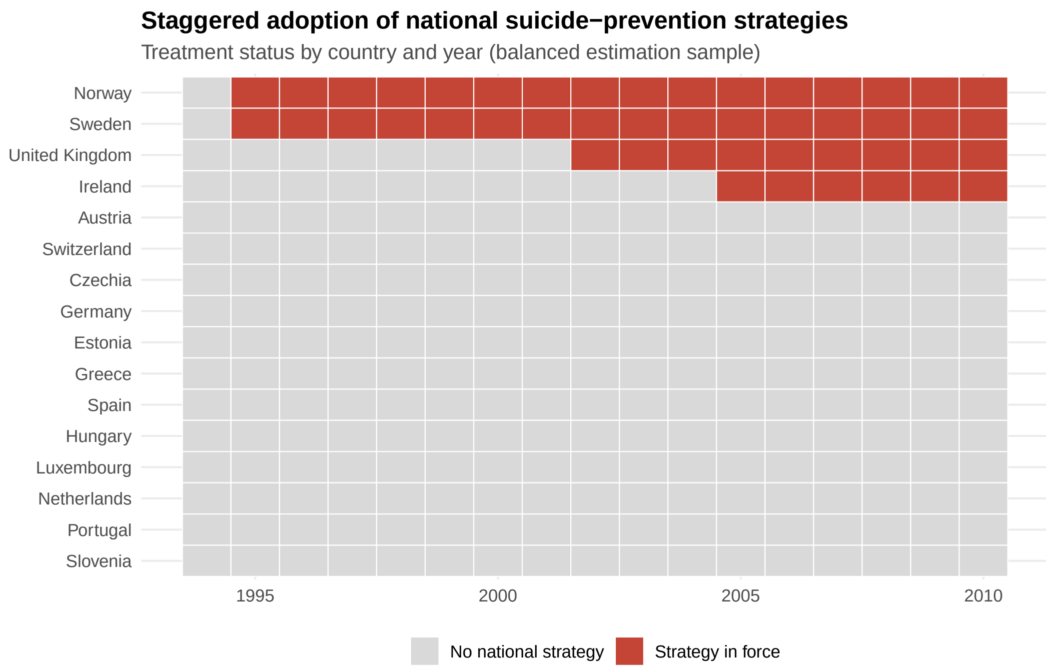

Figure 2: Staggered adoption of national suicide-prevention strategies, balanced estimation sample, 1994–2010. Red indicates a strategy in force.

stop here. Everything from this point forward is the argument that this number, however robust to the choices econometricians usually worry about, measures something other than the effect of the policy.

6 What the Number Is Not

It survives the pre-trend test—but the test is nearly blind. The event study in Figure 6 plots the Callaway–Sant'Anna effect by years since adoption, with every coefficient—before and after—a long difference from the year just before treatment, the same $g - 1$ baseline against which the post-adoption effects and the parallel-trends assumption itself are defined. Look left of zero. The pre-adoption coefficients are negative and noisy, and jointly they do not reject a flat path: the sum of squared t-statistics across the eight pre-periods is 9.9, comfortably inside what chance produces on eight degrees of freedom. Measured the right way, parallel trends cannot be rejected. I want to be exact about what that does and does not

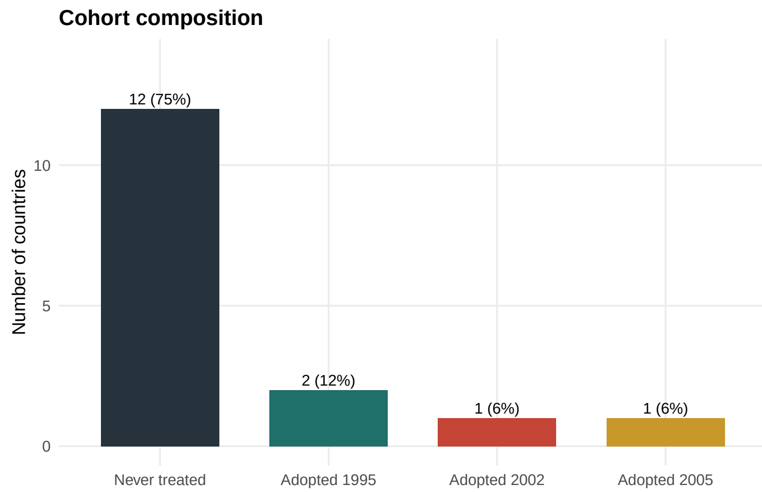

Figure 3: Cohort composition. Twelve never-treated controls; four treated countries in three adoption cohorts.

Table 1: The estimate is positive and robust to estimator choice

Estimator / specification	ATT (per 100,000)	SE
Callaway–Sant'Anna, reg., +unemployment (*primary*)	+2.73	1.05
simple average	+3.05	1.24
doubly robust	+2.56	—
unconditional	+2.76	—
not-yet-treated comparison	+2.66	—
Borusyak–Jaravel–Spiess (imputation)	+3.23	1.30
Sun–Abraham (interaction-weighted)	+3.03	1.38

buy. It removes the most common objection to a result like this—that the treated countries were already diverging—but it removes it the way an unlit room removes the objection that the furniture is ugly. With four treated countries the pre-trend test is powered to catch only gross violations; its silence is the silence of a design too small to hear anything, not a clean bill of health. (An earlier draft, using the short, period-to-period baseline that much applied work still reports, found a large spurious violation. The long-difference baseline is the correct one, and it is flat; the lesson that the choice of baseline can manufacture or erase a pre-trend is not the smallest one in this paper.)

The little the test can see, it sees through two countries. The pre-adoption leads are identified almost entirely off the United Kingdom and Ireland; Norway and Sweden, adopting

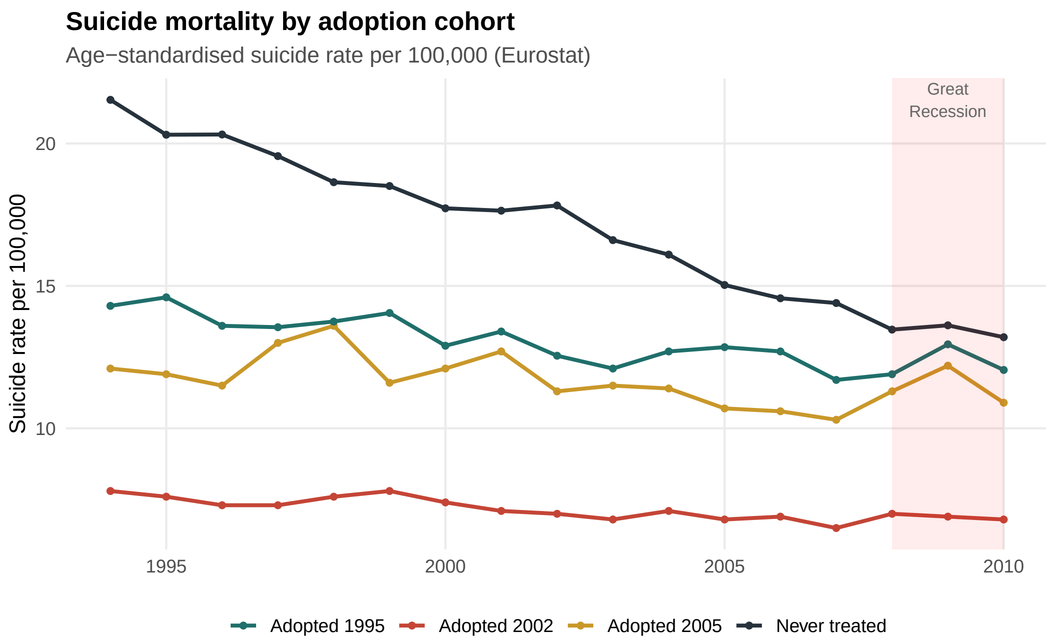

Figure 4: Suicide mortality by adoption cohort. The shaded band marks the 2008–2010 recession.

in 1995 with a single pre-period each, contribute to the *level* of the effect but nothing to the *test* of trends. The estimand leans hardest on the cohort that cannot be checked at all.

It is not free of the recession. Figure 7 re-aggregates the same group-time effects by calendar year rather than by event time. The estimated "effect" is small or negative through the mid-2000s and then leaps upward in 2008 and stays there. The treated cohorts' post-adoption windows—Ireland's especially, beginning in 2005—overlap precisely the years in which austerity and unemployment drove European suicide up [Reeves et al., 2012, Stuckler and Basu, 2013]. A staggered design cannot tell the effect of a 2005 policy apart from the effect of a 2008 recession that strikes the 2005 adopter during its post-period. The positive ATT may be, in substantial part, the recession wearing the policy's clothes; it may be selection; it may be noise. I do not claim to know which, and the claim is not that the recession explains the result—to assert that would be to identify, from the same panel I have just called uninformative, an effect I have no business identifying. The claim is the weaker and more durable one: that these two forces are present, that the design cannot separate

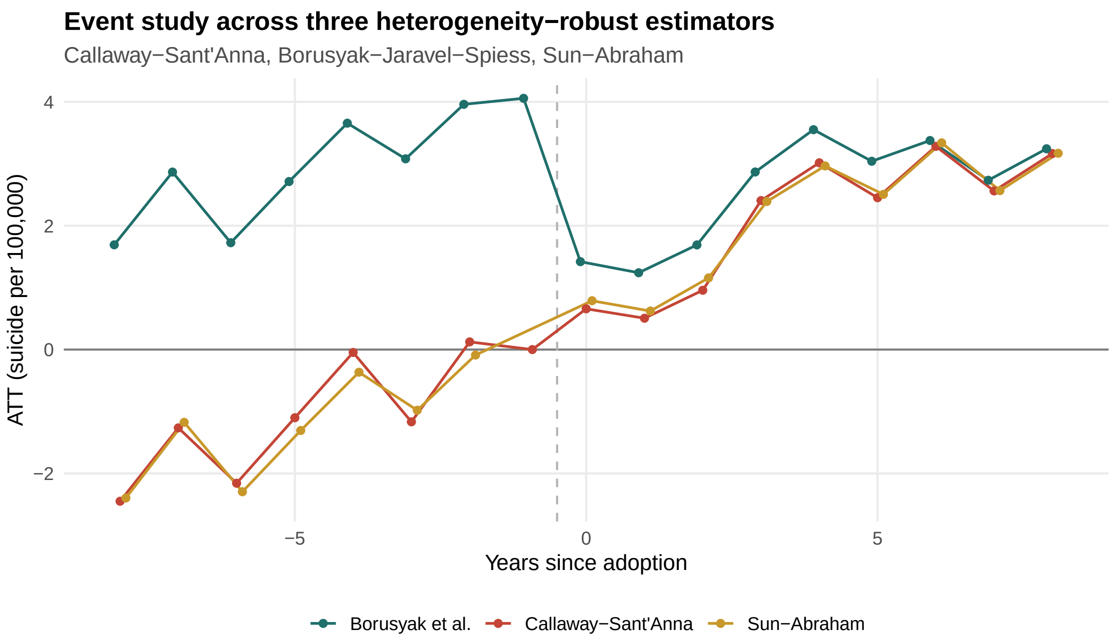

Figure 5: Event studies from three heterogeneity-robust estimators. The estimators agree.

them from the policy, and that a number which confounds all three is not a measurement of any one.

It cannot be signed. A pre-trend test that cannot reject is not the same as a parallel-trends assumption that holds; the honest question is how robust the estimate is to the violations the test is too weak to rule out. The sensitivity analysis of Rambachan and Roth [2023] answers it: allowing post-adoption deviations from parallel trends on the scale of the (noisy) pre-period wiggle already in the data, the robust confidence set for the effect includes zero, includes large increases, and includes large decreases. Figure 8 shows the set widening as the permitted violation grows; it crosses zero almost immediately. We cannot say from these data whether strategies raise suicide, lower it, or do nothing.

It is indistinguishable from chance. As a final check I reassign adoption at random among the never-treated controls, three hundred times, and re-estimate. If the true effect were real, the observed estimate should sit in the tail of this placebo distribution. It does not: a quarter of random placebos produce an estimate at least as large in magnitude (Figure 9).

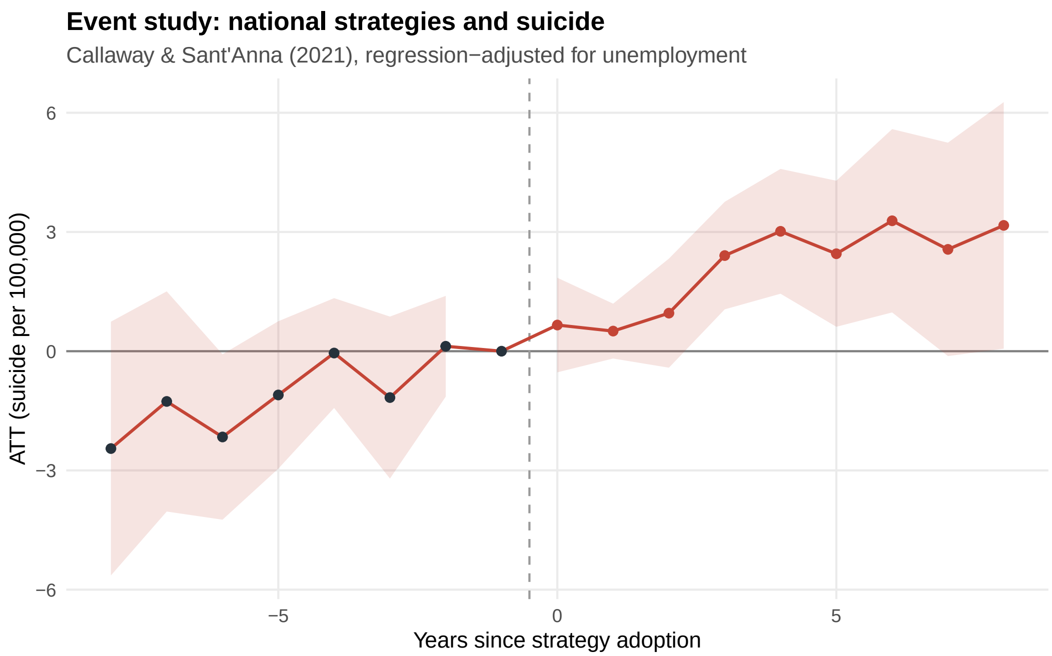

Figure 6: Event study, Callaway–Sant'Anna, universal (long-difference) baseline. Pre-adoption coefficients (dark) do not jointly reject parallel trends ($\sum z^2 = 9.9$, 8 d.f.); the post-adoption path is positive.

The observed +2.73 is an ordinary draw from a distribution generated by countries that never adopted anything.

Synthetic control agrees, where it can speak at all. The method built for a handful of treated units [Abadie et al., 2010, Ben-Michael et al., 2021] can be applied only to the two countries with enough pre-adoption history to construct a credible counterfactual—the United Kingdom and Ireland; Norway and Sweden, with a single pre-period each, defeat it entirely, which is itself a comment on how little the data contain. For both the UK and Ireland the augmented synthetic control reproduces the pre-adoption path well and then drifts above it after adoption, but the post-adoption gap is statistically ordinary: placebo p-values of 0.46 and 0.69 against control units treated as if they had adopted (Appendix B, Figures 10 and 11). Ireland's gap, the larger of the two, opens in 2008. Even the method of last resort returns the same verdict.

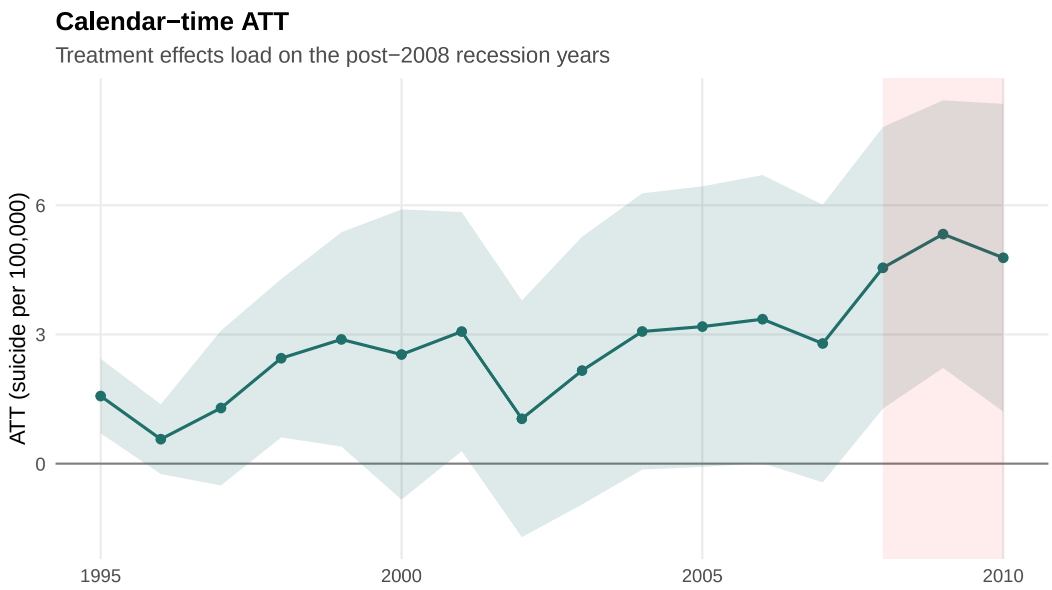

Figure 7: Calendar-time aggregation. The "effect" materializes in 2008–2010, the recession years.

It is fragile where it should be solid. A genuine treatment effect should not depend on which untreated countries one happens to compare against. This one does. Dropping the four post-communist controls—Czechia, Estonia, Hungary, Slovenia, whose suicide rates were falling steeply from post-transition peaks throughout the window—nearly halves the estimate, to +1.24 (standard error 0.76). Recoding Sweden's adoption to its 2008 parliamentary strategy leaves +2.41; allowing a year of anticipation leaves +1.68. The sign survives; the magnitude wanders with the specification, in the manner of an estimate that is measuring the comparison rather than the policy.

Five windows, one answer. They are not independent of one another, but they fail together, and a result that survives none of them is not a result. The cross-national mortality record cannot tell us what national suicide-prevention strategies do.

7 The Arithmetic of Governing in the Dark

It is tempting, having demolished an estimate, to feel that something has been accomplished. Very little has. The world's suicides are not reduced by a confidence interval, and a country

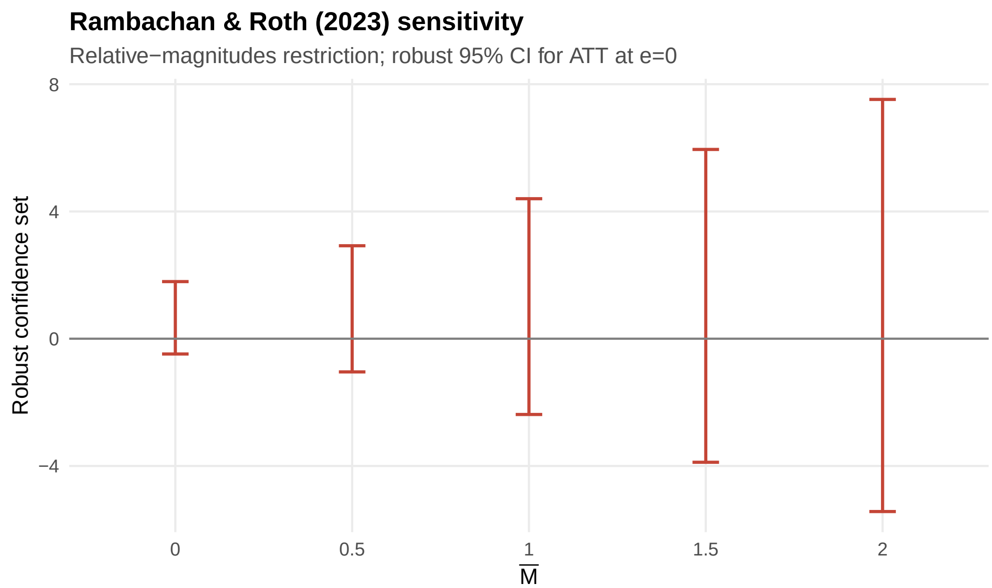

Figure 8: Rambachan–Roth sensitivity. Under modest departures from parallel trends, the robust confidence set for the post-adoption effect spans zero.

deciding tomorrow whether to write a strategy gains nothing from being told that the last twenty years of evidence were not what they seemed—unless that knowledge changes what it does next. So let me say what I think it should change.

It should change our confidence, not our conduct. The finding here is not that strategies are useless; it is that the observational cross-section is structurally incapable of revealing their worth, because countries adopt them in response to the very trends that would otherwise be used to identify their effect, and because the post-adoption years of the principal European adopters were swallowed by a recession that raised suicide everywhere. A policy can be both worth keeping and unproven, and intellectual honesty requires holding those two facts at once. What is not defensible is the laundering of association into causation that the field has practiced—the citing of a single pre-Goodman-Bacon comparison as though it settled the matter, the diffusion of a policy on the strength of evidence that evaporates the moment it is examined with the tools the rest of applied economics now considers mandatory.

It should also change how we invest in knowing. The reason we cannot answer the question is not that suicide is mysterious; it is that we adopted a continent's worth of

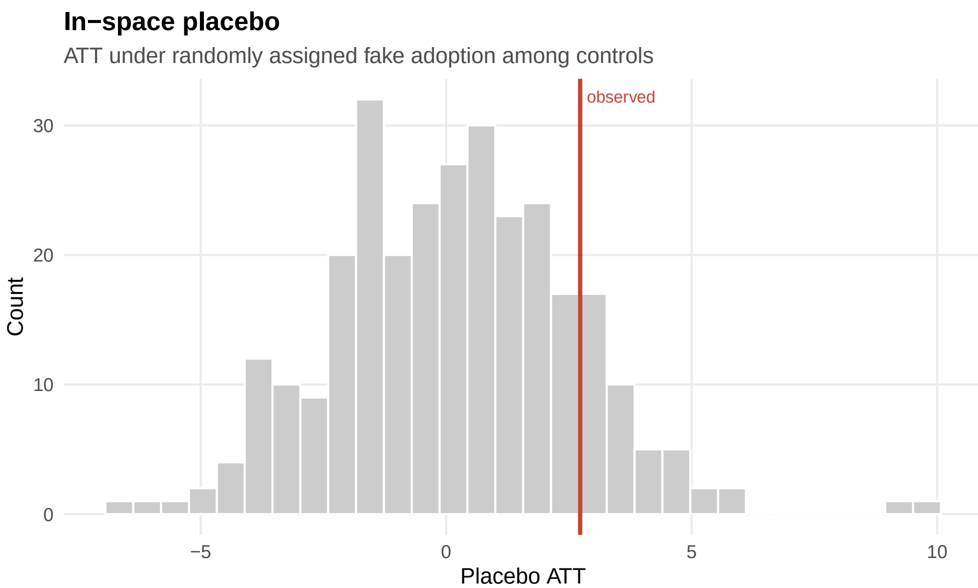

Figure 9: In-space placebo. The observed estimate (red line) is unremarkable against effects estimated from randomly assigned fake adoption among controls ($p = 0.25$).

strategies without once building the variation that would let us evaluate them. We have no staggered randomization, no phased regional rollout with administrative mortality data, no pre-registered evaluation attached to a single one of these national plans. Every country switched at once, at the level at which the confounders also live, in response to the outcome itself. If the European Commission wishes to know whether its mental-health investments save lives—and it should wish to know, because the alternative is to keep spending moral and fiscal capital on faith—then the evaluation must be designed before the policy is deployed, at a level finer than the nation, with the counterfactual built in rather than reconstructed afterward from a panel too small and too confounded to bear it.

I began with a number—one death every forty seconds—and I want to end with it, because it is the only number in this paper that is not in dispute. The strategies were written for that number. Whether they have moved it, I cannot tell you, and neither, I think, can anyone who looks honestly at the cross-national data. That should not comfort the skeptic or the believer. It should frighten both. We are making life-and-death policy in a darkness we have mistaken for evidence, and the first step out of it is to admit that the light we thought we had was never there.

A Appendix A: Data Construction and Treatment Coding

Outcome. Eurostat table `hlth_cd_asdr`, age-standardized death rate for intentional self-harm (ICD-10 X60–X84, Y87.0), sex total, per 100,000, 1994–2010 [Eurostat, 2026]. The regional table `hlth_cd_asdr2` (NUTS-2, 2011–2023) is not used for estimation because it postdates the adoption wave, but it confirms the national series in its overlap.

Treatment. A country is coded treated in the first year its government adopted a standalone national suicide-prevention strategy, following the inclusion criterion of the WHO's 2018 inventory [World Health Organization, 2018] and the peer-reviewed dates of Lewitzka et al. [2019]. In-window adopters: Norway 1995, Sweden 1995, United Kingdom 2002 (England's national strategy; Eurostat reports the UK in aggregate), Ireland 2005. Finland (1992) is dropped as already-treated at the panel's start; France (2000) is dropped because its outcome series begins in 2001, leaving no pre-period. Documented never-adopters used as controls include Germany, Italy, Spain, Belgium, Austria, Switzerland, the Netherlands, Portugal, Greece, Luxembourg, Czechia, Estonia, Hungary, and Slovenia; the balanced sample retains the twelve with complete outcome and covariate records (Belgium, Denmark, Italy, Poland, and Slovakia are dropped for gaps in the suicide series). The sign of the estimate survives recoding Sweden's adoption to its 2008 parliamentary strategy (+2.41) and permitting one year of anticipation (+1.68), but its magnitude nearly halves when the four post-communist controls are dropped (+1.24, s.e. 0.76). Denmark cannot be recoded as a 1998 adopter, contrary to what one might wish to test, because its suicide series is incomplete and it does not appear in the balanced panel; and adding real GDP per capita as a second covariate cannot be estimated at all, because the GDP-complete subsample is too small to support it—a limitation worth stating plainly rather than burying, and itself a measure of how little identifying information the data contain.

Covariate. National unemployment rate, World Bank modeled-ILO series `SL.UEM.TOTL.ZS`, 1994–2010 [World Bank, 2026]. Real GDP per capita and population age structure (Eurostat) are used in supplementary specifications.

Balance. Table 2 reports baseline (1994) means and the standardized difference in means—$(\bar{x}_T - \bar{x}_C)/\sqrt{(s_T^2 + s_C^2)/2}$—which Imbens and Rubin [2015] call "imbalanced" beyond 0.25 in absolute value. Every comparable covariate clears that bar. Unemployment, the covariate I condition on, is imbalanced at 0.33: the treated countries were not on the same cyclical footing as the controls, which is precisely why conditioning on it is necessary rather than cosmetic. The treated countries also begin with *much lower* suicide (−0.99) and an older age structure—a reminder that the cross-sectional contrast is uninformative and only the within-country trend comparison can be.

Reproducibility. All estimates were computed independently in R (`did`, `didimputation`, `fixest`, `HonestDiD`, `augsynth`), Python (`differences`), and Stata (`csdid`). Sample sizes and means are identical across languages; the primary ATT replicates to four to six significant

Table 2: Baseline (1994) balance, treated vs. never-treated. Standardized difference > 0.25 in absolute value is "imbalanced" [Imbens and Rubin, 2015].

Variable	Treated mean	Control mean	Std. diff.	Imbalanced?
Suicide rate (per 100k)	12.13	21.53	-0.99	yes
Unemployment rate (%)	9.78	8.18	$+0.33$	yes
Share aged 65+ (%)	15.23	13.95	$+0.63$	yes
Median age (years)	35.20	36.24	-0.42	yes

figures within matched specifications. The cross-language replication report is archived with the paper.

B Appendix B: Estimators, Synthetic Control, and Additional Robustness

Estimators. The primary estimates use Callaway and Sant'Anna [2021] with regression adjustment and the never-treated comparison group, aggregated to the group-size-weighted ATT and to event time. Borusyak et al. [2024] provides the imputation estimator and its long-horizon event study; Sun and Abraham [2021] provides the interaction-weighted estimator with never-treated as the reference cohort. Goodman-Bacon [2021] and de Chaisemartin and D'Haultfœuille [2020] motivate the abandonment of two-way fixed effects; Rambachan and Roth [2023] provides the sensitivity analysis. Doubly robust and inverse-propensity variants [Sant'Anna and Zhao, 2020] are reported but, given the absence of common support documented in Figure 1, are not preferred.

Augmented synthetic control. For the United Kingdom and Ireland I estimate a ridge-augmented synthetic control [Abadie et al., 2010, Ben-Michael et al., 2021] against the pool of never-treated donors, with placebo inference based on reassigning treatment to each donor in turn and ranking the post-to-pre RMSPE ratio. Figures 10 and 11 show the fit and the placebo distribution; the post-adoption gaps are statistically ordinary ($p = 0.46$ for the UK, $p = 0.69$ for Ireland) and, for Ireland, open with the 2008 recession. Norway and Sweden cannot be analyzed by synthetic control: with a single pre-treatment year each, no counterfactual can be constructed, which is itself a measure of how little identifying variation the 1995 cohort contains.

Leave-one-out. Dropping each treated country in turn leaves the sign and rough magnitude of the aggregate ATT unchanged; no single country drives the result, which is to say that the positive estimate is a feature of the whole confounded comparison and not an artifact of one outlier.

Is the two-way fixed-effects estimate the culprit? It is fair to ask whether the whole exercise is an artifact of the well-known pathology of two-way fixed effects under staggered adoption [Goodman-Bacon, 2021, de Chaisemartin and D'Haultfœuille, 2020]. It is not.

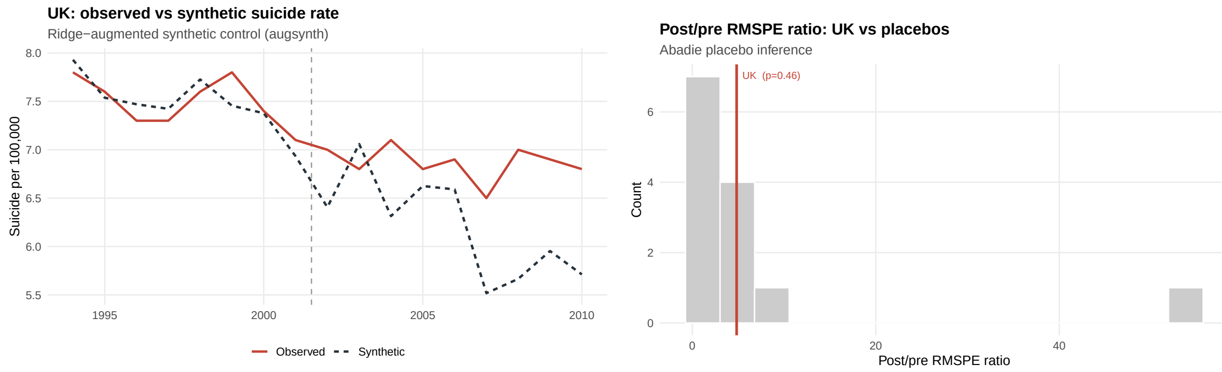

Figure 10: United Kingdom. Left: observed vs. synthetic suicide rate. Right: post/pre RMSPE ratio against control placebos ($p = 0.46$).

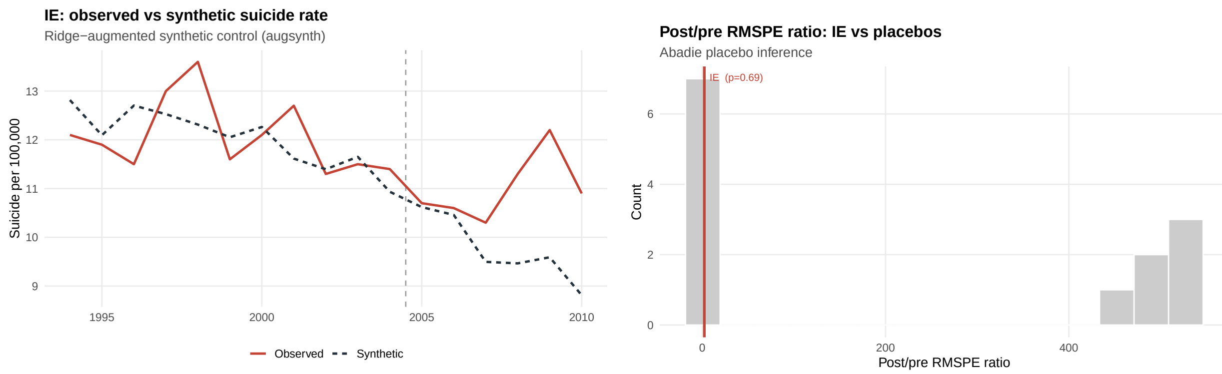

Figure 11: Ireland. Left: observed vs. synthetic suicide rate; the gap opens in 2008. Right: post/pre RMSPE ratio against control placebos ($p = 0.69$).

Running the static TWFE regression of suicide on a post-adoption indicator returns a coefficient of 3.11, and the Goodman-Bacon decomposition (Table 3) reconstructs it exactly as a weighted average of its 2×2 building blocks. The "forbidden" comparisons that use already-treated countries as controls—the source of the bias—carry only about three percent of the total weight, because the never-treated pool is large. The clean treated-versus-untreated comparisons carry the rest. TWFE (3.11) therefore sits right beside the heterogeneity-robust Callaway–Sant'Anna estimate (2.73): the estimator is not the problem. The problem is identification, and it is shared by every estimator in this paper.

Table 3: Goodman-Bacon decomposition of the TWFE estimate

2×2 comparison	Weight	Avg. estimate	Contribution
Treated vs. untreated	0.864	3.57	3.085
Later vs. earlier treated	0.112	0.29	0.032
Earlier vs. later treated	0.025	−0.22	−0.005
Reconstructed TWFE	1.000		3.112

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